Power BI Project Report
SUPERSTORE SALES ANALYSIS

1. Executive Summary

This project involved using Microsoft Power BI to transform a superstore retail sales dataset into an interactive business intelligence report. The project covered the required stages of importing the dataset into Power Query Editor, preparing the data, modelling the dataset, creating DAX measures, developing an interactive dashboard, deriving business insights and recommendations, importing data from MySQL into Power BI, and explaining Power BI Services.

The completed dashboard provides an overview of sales performance using cards, charts and interactive slicers. The analysis covers sales by state, region, product name, month, segment and category.

From the overall dashboard view, the business recorded approximately $2.26M in total sales, 5K orders, 793 customers and 2K products.

2. Project Requirements and Implementation

i. Import the datasets into Power Query Editor

The Superstore sales dataset was imported into Power Query Editor from Excel. It was use to prepare the data before it was loaded into Power BI Desktop for analysis and visualization.

ii. Clean the datasets

The dataset was reviewed and prepared for analysis. The fields and data types required for the analysis were checked so that they could be used correctly in calculations, relationships, visualizations and filters.

iii. Model the datasets

The prepared dataset was modelled in Power BI Desktop. A Date table was used to support time-based analysis and the DAX time-intelligence calculations.

All the column in the dataset were in the right format.

In modelling the dataset, a date table was created and used to create a relationship between the date and sales.

iv. DAX Measures

The project requirement specified a minimum of 15 DAX formulas. The following 15 measures were created:

a. Total Sales: calculates the total sales generated.
   Total sales = Sum(sales[sales])

b. Total Orders – calculates the number of unique orders.
   Total Order = Distinctcount(sales[order id])

c. Total Customers – calculates the number of unique customers.
   Total Customer = Distinctcount(sales[Customer id])

d. Total Products: calculates the number of unique products.
   Total Product = Distinctcount(sales[Product id])

e. Average Sales – calculates the average sales value.
   Average sales = Average(sales[sales])

f. Maximum Sale: identifies the highest individual sale.
   Maximum sales = Max(sales[sales])

g. Minimum Sale: identifies the lowest individual sale.
   Minimum sales = Min(sales[sales])

h. Average Sales Per Order: calculates the average sales value per order.
   Average Sales Per Order = Divide([Total Sales], [Total Order])

i. Sales YTD (year-to-date) : calculates sales from the beginning of the year to the selected date.
   Sales YTD = Total YTD([Total Sales], ‘Date’[Date])

j. Sales MTD (month-to-date : calculates sales from the beginning of the month to the selected date.
   Sales MTD = TotalMTD([Total Sales], ‘Date’[Date])

k. Sales Last Year: compares sales with the corresponding period from the previous year.
   Sales Last Year = Calculate[Total Sales], Sameperiodlastyear (‘Date’[Date])

l. Sales Growth: calculates the absolute change between current sales and previous-year sales.
   Sales Growth = [Total Sales] – [Sales last year]

m. Sales Growth %: calculates the percentage change in sales.
   Sales Growth % = Divide([Sales Growth], [Sales last year]

n. Sales Per Customer: calculates the average sales generated per customer.
   Sales Per Customer = Divide([Total Sales], [Total Customers])

o. Sales Per Product: calculates the average sales generated per product.
   Sales Per Product = Divide([Total Sales], [Total Products])

These measures provided the calculations needed to analyze business performance and create meaningful visualizations.

3. Dashboard Design

The final report was designed as a SUPERSTORE SALES DASHBOARD.

The dashboard contains four main cards:

Total Sales – $2.26M

Total Order – 5K

Total Customer – 793

Total Product – 2K

The dashboard also contains the following visualizations:

Top 5 Sales by State

Sales by Region

Top 5 Sales by Product Name

Monthly Sales

Sales by Segment

Sales by Category

To make the dashboard interactive, four slicers were added:

Year

Region

Segment

Category

These slicers allow users to filter the dashboard and examine sales performance from different perspectives.

4. Key Findings and Insights

Regional Performance

The West region recorded the highest sales among the four regions shown on the dashboard, followed by the East, Central and South regions.

This indicates that the West region is an important contributor to the company's overall sales performance.

State Performance

Among the Top 5 states displayed, California recorded the highest sales, followed by New York, Texas, Washington and Pennsylvania.

This shows that California is the strongest-performing state among the leading states in the analysis.

Category Performance

Technology recorded the highest sales among the three product categories, followed by Furniture and Office Supplies.

This indicates that Technology is an important revenue-generating category for the business.

Segment Performance

The Consumer segment recorded the highest sales, followed by Corporate and Home Office.

This suggests that Consumer customers represent the largest sales segment in the dataset.

Monthly Sales Performance

The monthly sales trend shows that sales fluctuate throughout the year. Some months recorded stronger performance while others experienced lower sales.

Analyzing these changes can help the business understand seasonal patterns and improve sales planning.

Product Performance

The Top 5 Product Name visual highlights the individual products that generated the highest sales.

5. Business Recommendations

Based on the findings from the Power BI analysis, the following recommendations can be made:

a. Maintain focus on Technology

Technology generated the highest sales among the categories analyzed. The business should continue to monitor this category and identify opportunities to maintain its strong performance.

b. Investigate strong regional performance

The business should investigate the factors contributing to the strong performance of the West region and leading states such as California. Successful strategies could potentially be applied to lower-performing regions.

c. Focus on the Consumer segment

Since the Consumer segment contributes the largest share of sales, the business should continue developing strategies that meet the needs of this customer group.

d. Monitor high-performing products

The business should regularly monitor its highest-performing products and ensure that inventory and promotional decisions are aligned with customer demand.

e. Analyze monthly sales patterns

The business should investigate the reasons behind periods of high and low sales. Understanding these patterns can help with seasonal planning and improve performance during weaker periods.

f. Use interactive filtering for decision-making

The Year, Region, Segment and Category slicers should be used to investigate specific areas of the business and support more focused decision-making.

6. MySQL Integration

The project instructions also required importing a table from a MySQL database into Power BI.

A connection was successfully established to the local MySQL server using the MySQL Connector/NET. The database used was employee_dataset.

The available tables included:

departments

employees

employees_log

performance

personnel

projects

salaries

training

The successful connection demonstrated Power BI's ability to connect to a relational MySQL database and bring database tables into the Power BI environment.

This also demonstrates how Power BI can work with data from different sources.

8. Power BI Services

Power BI Services is the cloud-based platform used to publish, share, manage and collaborate on Power BI reports and dashboards.

While Power BI Desktop is mainly used for connecting to data, transforming data, modelling data, creating DAX measures and designing reports, Power BI Service allows completed reports to be published and accessed online.

Some important features of Power BI Services include:

a. Publishing: Reports created in Power BI Desktop can be published to Power BI Service.

b. Sharing: Published reports and dashboards can be shared with authorised users.

c. Workspaces: Workspaces provide an environment for organising and managing Power BI reports, dashboards, datasets and other content.

d. Data Refresh: Power BI Service can be configured to refresh connected data sources so that reports can reflect updated information.

e. Security: Access permissions can be used to control who can view or work with reports and dashboards.

f. Mobile Access: Power BI reports and dashboards can also be accessed through supported mobile applications.

8. Conclusion

This project demonstrated the process of transforming a Superstore sales dataset into an interactive business intelligence solution using Microsoft Power BI.

The project covered Power Query data preparation, data modelling, DAX calculations, dashboard development, interactive slicers, business insights, recommendations and MySQL integration and Power bi Service.

The requirement for a minimum of 15 DAX formulas was fulfilled through measures covering sales, orders, customers, products, averages, time intelligence and sales growth.

The final dashboard provides a clear overview of business performance and allows users to interact with the data using Year, Region, Segment and Category slicers.

The analysis identified West region, California, Technology category and Consumer segment as important contributors to sales in the current dashboard view.

The MySQL integration also demonstrated Power BI's ability to connect to an external relational database, while the explanation of Power BI Services provided an understanding of how Power BI reports can be published, shared and managed after development.

Overall, the project demonstrates how Power BI can transform raw business data into meaningful visual insights that support data-driven decision-making.